

CHAPTER 6 : CLASSIFICATION OF PLANTS

Topic:-

Prepare an album of monocot and dicot seeds and display it in the classroom.

Introduction:-

Flowering plants, known scientifically as Angiosperms, represent the most advanced and diverse group in Kingdom Plantae. In these plants, reproductive seeds develop safely protected inside fruits. Based on the number of embryonic seed leaves—called **cotyledons**—present inside the seed, Angiosperms are broadly classified into two distinct classes: **Monocotyledonae (Monocots)** and **Dicotyledonae (Dicots)**.

A cotyledon serves as an embryonic food reserve that nourishes the developing seedling during the critical initial phases of germination until young green leaves emerge to conduct photosynthesis. Monocot seeds contain a single thin cotyledon (such as in maize, wheat, rice, and jowar), whereas dicot seeds possess two thick, fleshy cotyledons packed with nutrients (such as in gram, pea, groundnut, and beans).

To study and appreciate these morphological distinctions practically, our science team collected ten diverse grain and pulse seeds from local agriculture and kitchens. We soaked, dissected, preserved, and mounted them into a beautifully illustrated botanical seed album, which was displayed in our classroom for peer study.

❖ Key Distinctions Between Monocots and Dicots:-

- Monocots: 1 cotyledon, fibrous root system, parallel leaf venation, trimerous flowers.
- Dicots: 2 cotyledons, taproot system with lateral branches, reticulate leaf venation, pentamerous flowers.
- Germination: Monocots typically exhibit hypogeal germination; many dicots show epigeal germination.

Aim & Album Preparation Setup:-

Aim: To collect representative monocot and dicot seeds, study their embryonic morphology, prepare a durable transparent-pouch botanical seed album, and display it in the classroom.

❖ *Objectives of the Project:-*

- To collect five authentic varieties each of monocotyledonous and dicotyledonous seeds.
- To dissect soaked seeds to observe single versus dual cotyledon arrangements.
- To mount specimens neatly with scientific nomenclature, family, and culinary uses.
- To display the prepared album in our classroom science corner for interactive learning.

❖ *Materials & Tools Used:-*

- Seed samples: Maize, Wheat, Rice, Jowar, Bajra (Monocots); Bengal Gram, Pea, Bean, Groundnut, Mustard (Dicots).
- Sturdy scrapbook / ring binder with thick ivory mounting card sheets.
- Small self-sealing transparent zip pouches, double-sided adhesive tape, and label stickers.
- Petri dishes, forceps, dissecting needle, blotting paper, and magnifying hand lens (10x).



Step-by-Step Album Preparation:-

The compilation of the seed album was executed with scientific precision through the following five steps:

❖ *Album Fabrication & Mounting Procedure:-*

- **Step 1 (Sample Selection & Cleaning):** Procured healthy, undamaged, and pesticide-free seeds of 5 monocots (cereals/grains) and 5 dicots (pulses/oilseeds). Cleaned and air-dried all samples.
- **Step 2 (Soaking & Dissection):** Placed representative seeds of each type in moist petri dishes for 12 hours. Carefully peeled off the outer seed coat (testa) using forceps and split open dicots (like gram and pea) along the median groove to reveal the two distinct cotyledons and miniature embryo axis (radicle and plumule).
- **Step 3 (Specimen Preservation):** Sun-dried split specimens and intact whole seeds to prevent fungal or microbial growth before mounting.
- **Step 4 (Encapsulation & Layout):** Enclosed 5 to 6 clean seeds of each species into high-clarity transparent mini zip pouches. Firmly mounted them onto ivory album sheets using strong double-sided tape.
- **Step 5 (Botanical Labeling):** Inscribed detailed printed index cards below each pouch stating Common Name, Scientific Botanical Name, Family, Class, and Food Value.

Observations & Mounted Seed Catalog Table:-

The complete catalog of seeds preserved in our classroom botanical album is documented below:

Common Name	Botanical Name	Class	Cotyledons	Primary Food Reserve
Maize (Corn)	<i>Zea mays</i>	Monocot	Single (Scutellum)	Endosperm Starch
Wheat	<i>Triticum aestivum</i>	Monocot	Single	Carbohydrates & Gluten
Rice	<i>Oryza sativa</i>	Monocot	Single	Endosperm Starch
Jowar (Sorghum)	<i>Sorghum bicolor</i>	Monocot	Single	Starch & Minerals
Bengal Gram	<i>Cicer arietinum</i>	Dicot	Two Fleshy	High Protein & Iron
Green Pea	<i>Pisum sativum</i>	Dicot	Two Fleshy	Proteins & Vitamins
Groundnut	<i>Arachis hypogaea</i>	Dicot	Two Fleshy	Vegetable Lipids/Proteins

Every monocot sample resisted splitting, while every dicot sample separated effortlessly into two equal symmetrical halves.

Scientific Analysis & Anatomical Insights:-

Detailed structural examination of the soaked seeds reveals fundamental evolutionary and developmental biology principles:

❖ *Anatomical & Physiological Insights:-*

- **Endospermic vs Non-Endospermic Seeds:** In monocot seeds (like maize and wheat), food is stored in a large, distinct endosperm tissue separated from the single cotyledon (scutellum) by an aleurone layer. In contrast, in most dicots (like gram and pea), the endosperm is completely absorbed during seed development, and food is stored directly within the two fleshy cotyledons.
- **Embryo Axis Architecture:** The embryo axis consists of the radicle (which develops downward into the root) and the plumule (which develops upward into the shoot). In monocots, protective sheaths shield these tips—the coleorhiza covers the radicle and the coleoptile covers the plumule.
- **Evolutionary Significance:** Botanists recognize that dicotyledons represent the ancestral lineage of Angiosperms, from which specialized monocotyledons evolved with adaptations suited for grasslands and efficient wind-pollinated cereal reproduction.

Classroom Display & Educational Value:-

Our seed album was mounted in the science exhibition corner of our classroom, serving as an interactive learning aid:

❖ Classroom Exhibition & Peer Feedback:-

- **Interactive Sensory Learning:** Classmates could examine both intact and split seeds up close under hand lenses, observing the embryonic plumule and radicle without damaging delicate specimens.
- **Agricultural & Nutritional Context:** Monocots supply staple dietary carbohydrates (cereals that provide 60% of human caloric intake globally), while dicots supply vital plant proteins (pulses) and healthy edible oils.
- **Commendation by Science Faculty:** Our subject teacher praised the clean, transparent packaging that protects samples from moisture and insects, recommending it as a permanent classroom teaching aid.

❖ Album Maintenance Tips:-

- Keep the album in a dry, well-ventilated cabinet away from direct sunlight to prevent seed discoloration.
- Add dried neem leaves or silica gel pouches inside the binder jacket to repel stored-grain pests.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Mastered the practical botanical identification and classification of monocot and dicot seeds.
- Learned proper techniques of biological dissection, dehydration, and scientific herbarium mounting.
- Appreciated the fundamental role of seed morphology in agricultural science and human nutrition.

❖ Project Conclusion:-

Our seed album successfully demonstrated the structural contrasts between monocotyledonous and dicotyledonous flowering plants. By displaying ten neatly labeled specimens alongside anatomical diagrams, the project provided a valuable hands-on educational resource for our classroom. This exercise proved that tangible biological collections make taxonomy vivid and memorable.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 6: Classification of Plants).
- Bhojwani, S. S. & Bhatnagar, S. P. – *The Embryology of Angiosperms*, Vikas Publishing House.
- Indian Council of Agricultural Research (ICAR) – Handbook of Agriculture and Seed Technology.